

# Homework 3 Solutions

## Problem 1

Maya is a basketball player who makes 40% of her three point field goal attempts. Suppose that at the end of every practice session, she attempts three pointers until she makes one and then stops. Let  $X$  be the total number of shots she attempts in a practice session. Assume shot attempts are independent, each with probability of 0.4 of being successful.

1. Identify by name the distribution of  $X$ . Be sure to specify the values of any relevant parameters.
2. Specify the probability mass function of  $X$ . Be sure to specify the possible values.
3. Construct a table, plot, and spinner corresponding to the distribution of  $X$ .
4. Compute  $P(X > 5)$  without summing.
5. Compute and interpret  $E(X)$ .
6. Compute and interpret  $SD(X)$ .

## Solution

1.  $X$  counts the number of successes until the first success in a sequence of Bernoulli(0.4) trials, so  $X$  has a Geometric(0.4) distribution.
2.  $X$  can take values 1, 2, 3, .... Even though it is unlikely that  $X$  is very large, there is no fixed upper bound. Even though  $X$  can take infinitely many values,  $X$  is a discrete random variable because it takes *countably* many possible values. In order for  $X$  to take value  $x$ , the first success must occur on attempt  $x$ , so the first  $x - 1$  attempts must be failures.

$$p_X(x) = (1 - 0.4)^{x-1}(0.4), \quad x = 1, 2, 3, \dots$$

3. See below.  $P(X = 1) = 0.4$ . If Maya does this every practice, then in about 40% of practices she will make her first three pointer on her first attempt.  $P(X = 2) = (0.6)(0.4) = 0.24$ . If Maya does this every practice, then in about 24% of practices she will make her first three pointer on her second attempt. And so on.

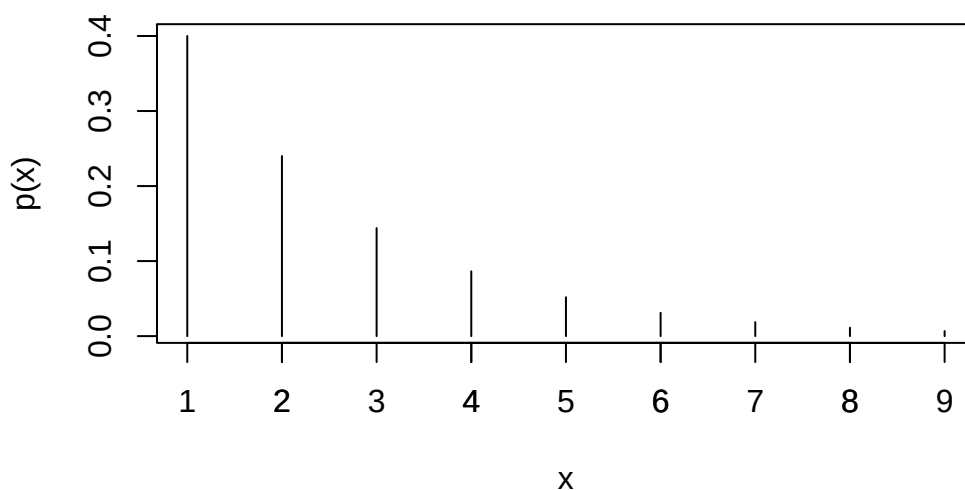
4. The key is to realize that Maya requires more than 5 attempts to obtain her first success if and only if the first 5 attempts are failures. Therefore,

$$P(X > 5) = (1 - 0.4)^5 = 0.078$$

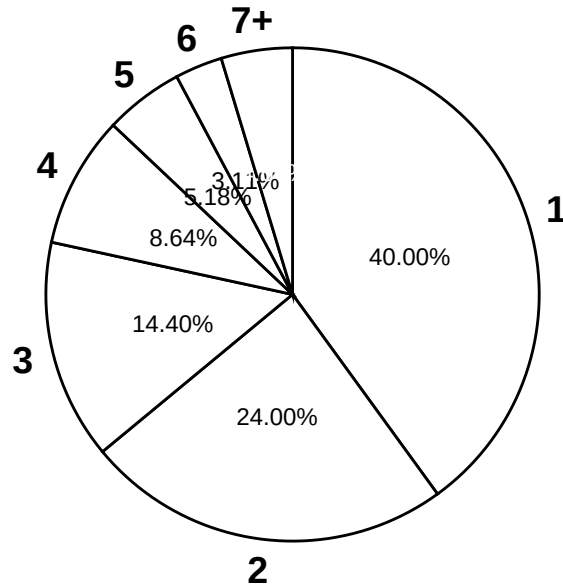
5. Since  $X$  has a Geometric(0.4) distribution,  $E(X) = 1/0.4 = 2.5$ . Over many practice sessions, she attempts on average 2.5 three pointers until she makes one per practice.
6. Since  $X$  has a Geometric(0.4) distribution,  $\text{Var}(X) = (1 - 0.4)/0.4^2 = 3.75$  so SD is 1.94. Over many practice sessions, the number of attempts until making one varies by about 1.94 attempts per practice on average.

| x | p(x)   |
|---|--------|
| 1 | 0.4000 |
| 2 | 0.2400 |
| 3 | 0.1440 |
| 4 | 0.0864 |
| 5 | 0.0518 |
| 6 | 0.0311 |
| 7 | 0.0187 |
| 8 | 0.0112 |
| 9 | 0.0067 |

**Geometric(0.4) pmf**



Geometric(0.4) distribution



## Problem 2

Ron and Leslie agree to the following bet. They'll ask Professor Ross if he saw the Eras Tour live. If he did, Leslie will pay Ron \$200; if not, Ron will pay Leslie \$100. (Neither has any direct information about whether or not Prof Ross saw the Eras Tour live.)

1. Given this setup, which of the following is being judged as more likely: that Prof Ross saw the Eras Tour, or that he did not? Why?
2. Ron and Leslie agree that this is a fair bet, and neither would accept worse odds. What is their subjective probability that Professor Ross saw the Eras Tour?
3. Suppose they were to hypothetically repeat this bet many times, say 3000 times. Given the probability from the previous part, how many times would you expect Leslie to win? To lose? What would you expect Leslie's net dollar winnings to be? In what sense is this bet "fair"?
4. Let  $X$  be Leslie's net winnings. Identify the distribution of  $X$ . (Remember: Leslie's winnings are Ron's losses and vice versa.)
5. Compute and interpret  $E(X)$ .

## Solution

1. The larger potential payout corresponds to the *less* likely event. So they think Professor Ross is more likely to *not* have a TikTok account than to have one.
2. The payouts are in a 2 to 1 ratio, so the odds that Professor Ross has a TikTok account are 2 to 1 against. So Professor Ross is twice as likely to not have a TikTok account than to have one. This corresponds to a subjective probability<sup>6</sup> that Professor Ross has a TikTok account of  $1/3$  (and a probability that he does not have one of  $2/3$ ).
3. The probability that Leslie wins is  $2/3$ , so you would expect her to win in 2000 of the 3000 repetitions. She wins \$100 each time she wins, so you would expect her to win a total of \$200,000 on games she wins. The probability that she loses is  $1/3$ , so you would expect her to lose in 1000 of the 3000 repetitions. She loses \$200 each time, so you would expect her to lose a total of \$200,000 on the games she loses. So you would expect Leslie's net winnings to be 0, and likewise for Ron.

| Win-<br>ner | Expected number of<br>repetitions | Leslie's winnings per<br>repetition (\$) | Leslie's expected total<br>winnings (\$) |
|-------------|-----------------------------------|------------------------------------------|------------------------------------------|
| Leslie      | 2000                              | 100                                      | 200,000                                  |
| Ron         | 1000                              | -200                                     | -200,000                                 |
| Total       | 3000                              | NA                                       | 0                                        |

4. The bet is fair in the sense that neither party is expected to profit or lose in the long run.
5.  $X$  is 100 with probability  $2/3$  and -200 with probability  $1/3$ , so  $E(X) = (100)(2/3) + (-200)(1/3) = 0$ . Over many such bets, Leslie's net winnings per bet would be 0 on average.

## Problem 3

Suppose that a total of 350 students at a college are taking a particular statistics course. The college offers five sections of the course, each taught by a different instructor. The class sizes are shown in the following table.

| Section | A | B | C | D | E |
|---------|---|---|---|---|---|
|---------|---|---|---|---|---|

<sup>6</sup>Technically, Ron and Leslie could still have different subjective probabilities. Leslie would not agree to worse odds, but she would accept better if Ron offered them. For example, given a potential loss of \$200, Leslie would also agree to a potential payout from Ron of \$125 rather than \$100. That is, Leslie would accept odds of 1.6 to 1 against ( $200/125 = 1.6$ ), corresponding to a subjective probability of  $1/(1 + 1.6) = 0.385$ . So Leslie's subjective probability that Professor Ross has a TikTok account is *at least*  $1/3$ . Similarly, Ron's subjective probability that Professor Ross has a TikTok account is *at most*  $1/3$ .

|                    |    |    |    |    |     |
|--------------------|----|----|----|----|-----|
| Number of students | 35 | 35 | 35 | 35 | 210 |
|--------------------|----|----|----|----|-----|

We are interested in: What is the average class size?

1. Suppose we randomly select one of the 5 *instructors*. Let  $X$  be the class size for the selected instructor. Specify the distribution of  $X$ . (A table is fine.)
2. Compute and interpret  $E(X)$ .
3. Compute and interpret  $P(X = E(X))$ .
4. Suppose we randomly select one of the 350 *students*. Let  $Y$  be the class size for the selected student. Specify the distribution of  $Y$ . (A table is fine.)
5. Compute and interpret  $E(Y)$ .
6. Compute and interpret  $P(Y = E(Y))$ .
7. Comment on how these two expected values compare, and explain why they differ as they do. Which average would you say is more relevant?

### Solution

1.  $X$  takes the value 210 with probability  $1/5$  and 35 with probability  $4/5$ .
2.  $E(X) = 210(1/5) + 35(4/5) = 70$  is the average number of students per class from the instructors' perspective. If we randomly select an instructor, record the number of students in the instructor's class, and repeat, the long run average number of students will be 70.
3.  $P(X = E(X)) = P(X = 70) = 0$ . No instructor has a class whose size is equal to the average class size (from the instructor's perspective).
4.  $Y$  takes the value 210 with probability  $210/350$  and 35 with probability  $140/350$ .
5.  $E(Y) = 210(210/350) + 35(140/350) = 140$  is the average number of students per class from the students' perspective. If we randomly select a student, record the number of students in the selected student's class, and repeat, the long run average number of students will be 140.
6.  $P(Y = E(Y)) = P(Y = 140) = 0$ . No student is in a class whose size is equal to the average class size (from the students' perspective).
7. Colleges usually report average class size from the instructor/class perspective, which in this case would be 70 students. From the students' perspective, the average class size is not even close to 70! In fact, it's twice that size. Some students (140 of them, which is 40% of the total of 350 students) have the benefit of a small class size of 35. But most students (210 of them, which is 60% of the students) are stuck in a large class of 210 students. In other words, most students would be pretty seriously misled if they chose this college based on the advertised average class size of 35 students per class. From the students' perspective, it seems that 140 is the more relevant average to report. However, neither average really adequately represents what is happening here: there is wide variability in class sizes between large and small.

## Problem 4

(This is an overly simplified example from actuarial science.) A life insurance company sells a term insurance policy to a 21 year old man (the “insured”) that pays \$100,000 to a designated beneficiary if the insured dies within the next 5 years, and pays nothing if the insured does not die before age 26. The probability that a randomly chosen 21-year-old man will die each year can be found in mortality tables. The company collects a premium of \$250 at the beginning of each year of the 5 years of the term, as long as the insured is alive, as payment for the insurance. The amount  $Y$  (in dollars) that the company earns on this policy is \$250 per year, less the \$100,000 that it pays out in the event the insured dies. The company doesn’t pay out anything if the insured does not die in the 5 year term.

| Age at death | Earnings $Y$ | Probability |
|--------------|--------------|-------------|
| 21           | −99,750      | 0.00183     |
| 22           |              | 0.00186     |
| 23           |              | 0.00189     |
| 24           |              | 0.00191     |
| 25           |              | 0.00193     |
| 26 or older  |              |             |

1. Find the distribution of  $Y$  by completing the above table. For example, the value -99,750 provides an example of how  $Y$  is calculated when the insured dies in the first year of the policy.
2. Compute  $E(Y)$ .
3. Write a sentence explaining in this context in what sense the number from the previous part is “expected”.
4. Would you be willing to sell such an insurance policy to one of your 21 year old friends? Explain why this is good business for the insurance company, but not for you.
5. Compute  $\text{Var}(Y)$
6. Compute  $\text{SD}(Y)$ .
7. Compute the probability that  $Y$  is more than 1 standard deviation above its mean.

## Solution

| Age at death | Earnings $Y$ | Probability |
|--------------|--------------|-------------|
| 21           | 99,750       | 0.00183     |
| 22           | -99,500      | 0.00186     |
| 23           | -99,250      | 0.00189     |
| 24           | -99,000      | 0.00191     |
| 25           | -98,750      | 0.00193     |

| Age at death | Earnings $Y$ | Probability |
|--------------|--------------|-------------|
| 26 or older  | 1,250        | 0.99058     |

1. See above. The earnings are computed according to the contract; the insurance company only needs to pay out the 100000 if the insured dies before age 26. Since all the probabilities must add up to 1, the probability in the blank is  $1 - (0.00183 + 0.00186 + 0.00189 + 0.00191 + 0.00193) = 0.99058$
2.  $E(Y) = (-99750)(0.00183) + \dots + (-98750)(0.00193) + (1250)(0.99058) = 303.23$
3. If the insurance company sells a large number of these policies to 21 year-old men then the insurance company will expect to make, on average, a profit of \$303.36 per policy. For example, if they sell 1 million policies, their total profit would be expected to be close to \$303 million.
4. If you sell one policy, you are either going to make a profit of \$1250, or you will pay out about \$100,000. The chances are high you will make a profit of \$1250, but there is a risk that you will have to pay out around \$100,000. This risk is probably too great to take on considering the relatively small (in comparison) reward.

On the other hand, the insurance company sells *many* policies. They operate in the long run. Over many policies they make a profit, on average, of about \$300 per policy. For example, for a million policies this equates to a profit of \$300 million. There would be some variability around this, but in the long run the average will win out. The company could be sure that they would only pay out the \$100,000 benefit for less than 1% of policyholders, and they will be earning \$1,250 on many other policies, more than enough to offset the relatively few \$100,000 payouts there are.

5.  $E(Y^2) = (-99750)^2(0.00183) + \dots + (-98750)^2(0.00193) + (1250)^2(0.99058) = 94328849$ . Then  $\text{Var}(Y) = E(Y^2) - (E(Y))^2 = 94328849 - (303)^2 = 94236827$ . This is a giant number because it's measured in dollars<sup>2</sup>.
6.  $\text{SD}(Y) = \sqrt{\text{Var}(Y)} = 9707$  dollars.
7. 0. One SD above the mean here is  $303 + 9707 = 10000$  and  $P(Y > 10000) = 0$  since it's not possible to have a value of this  $Y$  which is more than 1 SD above its mean.

## Problem 5

In a sample of 50 Olympic women's gymnasts the heights have a symmetric bell-shaped distribution with mean 60 inches and standard deviation 3 inches. In a sample of 50 Olympic men's basketball players the heights have symmetric bell-shaped distribution with mean 76 inches and the standard deviation 5 inches. Suppose the two samples are combined to obtain a sample with 100 heights. Choose one of the options below to complete the following sentence;

explain your reasoning without doing any calculations (hint: drawing a picture might help; also, what is the mean of the combined sample?). The standard deviation of the combined sample is:

- a. Less than 3 inches
- b. Equal to 3 inches
- c. Between 3 and 5 inches
- d. Equal to 5 inches
- e. Greater than 5 inches

### **Solution**

Greater than 5 inches. As a very rough measure of variability, think about the overall range. The range of heights for both groups combined would be greater than the range of heights for either of the two groups separately, so you would expect more variability when the groups are combined. To see what happens with SD, it's easiest to draw a picture. Looking at the gymnasts alone, the SD of 3 is the average distance from the mean of 60 inches. Looking at the men's bball players alone, the SD of 5 is the average distance from the mean of 76. If the two samples are combined, the sample mean height would be 68 inches (right in the middle of 60 and 76 because the groups have the same size). The SD in the combined sample measures the average distance from the mean of 68. The gymnasts heights tend to be closer to their mean of 60 than the overall mean of 68; similarly, the men's bball player heights tend to be closer to their mean of 76 than the overall mean of 68. So the average distance from the mean will be greater for the combined sample than for either of the two separate samples.